

circle.

11 D At point P,

$$\begin{aligned}\Phi_P &= \left(-\frac{GM}{r/3}\right) + \left(-\frac{GM_B}{2r/3}\right) \\ &= -\frac{G}{r}(3M + 1.5M_B)\end{aligned}$$

At point Q,

$$\begin{aligned}\Phi_Q &= \left(-\frac{GM}{2r/3}\right) + \left(-\frac{GM_B}{r/3}\right) \\ &= -\frac{G}{r}(1.5M + 3M_B)\end{aligned}$$

$$\Phi_Q = 1.25\Phi_P, \text{ so}$$

$$(1.5M + 3M_B) = 1.25(3M + 1.5M_B)$$

$$\Rightarrow M_B = 2M$$

12 D Consider the forces on the satellite: